

Michael Clayton

Sr. Python Full Stack Developer

✉ michaelclayton@gmail.com ☎ (915) 799 0333 📍 Astoria, NY

Summary

- Over **10+** years of IT experience in designing, developing, and implementing enterprise applications using **Python**, **Django**, and various technologies.
- Proficient in developing web applications with **Django**, **Flask**, and **RESTful APIs**, utilizing Python libraries such as **Beautiful Soup**, **NumPy**, **Pandas**, and **Flask libraries**.
- Skilled in front-end development using **ReactJS**, **NodeJS**, **Bootstrap**, **JavaScript**, **HTML**, **CSS**, **JSON** and **jQuery** with knowledge of W3C standards.
- Proficient in developing multi-threaded web services using **Cherry PY** & **bottle PY framework**.
- Developed and deployed machine learning models for predictive analytics, **NLP**, **image recognition**, and **recommendation systems** using frameworks like **TensorFlow**, **PyTorch**, and **scikit-learn**.
- Leveraged cloud platforms, **MLOps** workflows, and big data tools to optimize model training, deployment, and performance evaluation in production environments.
- Strong understanding of **DevOps** practices, including continuous integration and deployment (**CI/CD**) with tools like **Jenkins** and **Heroku**.
- Knowledgeable in containerization with **Docker**, configuration management with **SaltStack** and **Puppet**, and working with queueing architectures like **RabbitMQ**.
- Experienced in developing applications using **Java API**, **Swing**, **HTML 5**, **Flex**, **JavaScript**, **TypeScript**, **CSS3**, **jQuery**, **JDBC**, **JSF**, **JMS**, and **XML**.
- Experience building **cloud-native** applications in cloud environments such as **AWS** and **Azure**.
- Experienced in **NoSQL** technologies like **MongoDB**, **Couch DB** **Cassandra**, **Redis** and **relational databases** like **Oracle**, **SQLite**, PostgreSQL, and MySQL databases .
- Experienced with **automated deployment** tools like **Git** and **version control systems** such as Git.
- Collaborated closely with cross-functional teams in **Agile** environments to ensure seamless integration and deliver high-quality, user-focused solutions.

Skills

Languages

Python, Java, C/C++, Perl, Ruby

Markup Languages

HTML, CSS, JS/TS, XML, JSON

Scripting Languages

Java Script, TypeScript, CSS, AJAX, jQuery, Shell Scripting

Methodologies & Tools

Object Oriented Programming, UML, Agile Scrum

Databases

My SQL, No SQL, Apache Cassandra, Apache Airflow ,MongoDB, PostgreSQL, MS-SQL, Oracle

Servers

Apache Tomcat, Restful Web Services, IBM Web sphere, Open Stock

Frameworks

Django, Flask, Python, Pyramid, Cherry PY, PySpark

Education

New York City College of Technology
Bachelor's Degree in Computer Science

2010 – 2014
NY

Professional Experience

Senior Python Full Stack Engineer

MD THINK (Aug. 2020 ~ Dec. 2024)

- Contributed to the development of the **Integrated Data Management System**, a comprehensive platform for managing data pipelines, analytics, and applications, aimed at enhancing data accessibility and operational efficiency across multiple teams.
- Developed scalable web applications and APIs using **Python**, **Django**, and modern frontend technologies like **React**, **HTML5**, and **CSS**, ensuring seamless integration with backend services and responsive design principles.
- **Automated data scraping** and retrieval workflows using Python libraries such as requests and urllib, significantly reducing manual data collection time.
- Containerized and orchestrated microservices using **Docker** and **Kubernetes**, enabling high availability, streamlined deployments, and simplified maintenance.
- Built dynamic front-end features using React.js, enhancing user experience through interactive components, optimized state management, and modern UI design.
- Streamlined repetitive tasks and workflows with a **custom automation framework** built using **Python** and **Shell scripting**, saving development hours and reducing errors.
- Applied advanced **machine learning techniques** using **PySpark MLlib** for feature extraction, model training, and evaluation, improving predictive analytics capabilities.
- Set up and maintained **CI/CD pipelines** using **Jenkins** and **Git**, ensuring faster, more reliable builds and deployments across development and production environments.
- Designed and optimized **Snowflake data models**, leveraging materialized views to enhance performance and scalability for analytics workloads.
- Automated complex **ETL** workflows using **Apache Airflow**, reducing processing times and increasing system reliability.
- Collaborated with cross-functional teams to build scalable data pipelines, optimizing big data processing for real-time and batch workloads.
- Led the migration of legacy ETL processes to Apache Airflow, reducing maintenance overhead by 30% and improving scalability.
- Optimized SQL queries and developed robust data models for relational databases like **PostgreSQL** and **MySQL**, ensuring efficient data retrieval and storage.
- Created interactive Tableau dashboards to visualize key performance indicators and data trends, empowering stakeholders to make data-driven decisions.
- Deployed and managed **cloud-based applications** on **AWS** and **Google App Engine**, leveraging cloud-native solutions to enhance scalability and reduce costs.
- Applied **Agile** methodologies throughout the development lifecycle, including sprint planning, daily stand-ups, and retrospectives, ensuring timely delivery of high-quality features.
- Conducted comprehensive unit and integration testing using the **Pytest** framework, achieving 95% test coverage and minimizing production bugs.
- Managed version control and seamless deployments using **Git** and **Heroku**, ensuring smooth collaboration and continuous delivery.

Python Full Stack Developer

IWP Capital (Dec. 2018 ~ Jun. 2020)

- Played a key role at IWP Capital, contributing to the creation of cutting-edge financial technology solutions to streamline data handling and enhance client services.
- Developed full-stack web applications using **Python** and **Django** for backend processing and **ReactJS** for an interactive and intuitive user interface.
- Built efficient and secure **APIs** to enable smooth client-server communication, ensuring high reliability and performance.
- Worked on database design and optimization, leveraging **PostgreSQL** and **MongoDB** to ensure scalability and data consistency.
- Streamlined data workflows by integrating **Apache Airflow** for automated task scheduling and monitoring, significantly reducing manual intervention.
- Conducted rigorous data validation within **ETL** pipelines, ensuring accuracy and consistency between source and destination systems.
- Created modular **Python** scripts for data manipulation, CRUD operations, and complex mathematical calculations to support business processes.
- Deployed applications and services on **AWS**, utilizing EC2 for compute needs and S3 for scalable storage solutions.
- **Automated infrastructure provisioning** using **Ansible** and **CloudFormation** templates, reducing setup time and minimizing configuration errors.
- Designed and implemented scalable and fault-tolerant systems using AWS components such as **Route53**, **RDS**, **DynamoDB**, and **SQS**.
- Conducted extensive **unit** and **integration testing**, ensuring the stability and robustness of web applications and services.
- Used **Git** for version control and collaborated on deployment workflows to streamline development and minimize downtime.
- Participated in **Agile** ceremonies, contributing to sprint planning, retrospectives, and continuous delivery improvements.

Python Developer

IDEXX (Oct. 2014 ~ Oct. 2018)

- Automated data analysis tools and scripts to enhance the evaluation of diagnostic and laboratory data for accuracy and integrity.
- Designed and developed scalable enterprise web applications leveraging **Python** and **Django**, with a focus on improving diagnostic workflows and data accessibility.
- Designed intuitive user interfaces for internal diagnostic tools and customer-facing platforms using **HTML**, **CSS**, **JavaScript**, and **AJAX**, ensuring a seamless user experience for laboratory professionals.
- Parsed and validated diagnostic data by developing Python scripts to process large XML files and load results into structured **databases** for further analysis.
- Migrated on-premises ETL pipelines from **MS SQL Server** to **Azure Cloud** using **Azure Data Factory** and **Databricks**, enabling real-time analytics on diagnostic data.
- Created Python scripts to load, transform, and store healthcare data in **AWS**, **Azure**, and **GCP** cloud environments for improved scalability and security.
- Used **jQuery** to manage and enhance client-side scripting for diagnostic applications.
- Debugged and troubleshooted web applications to ensure optimal performance and reliability in handling sensitive diagnostic data.
- Utilized **Git** for version control, ensuring efficient collaboration and automated deployment of updates to laboratory management systems.